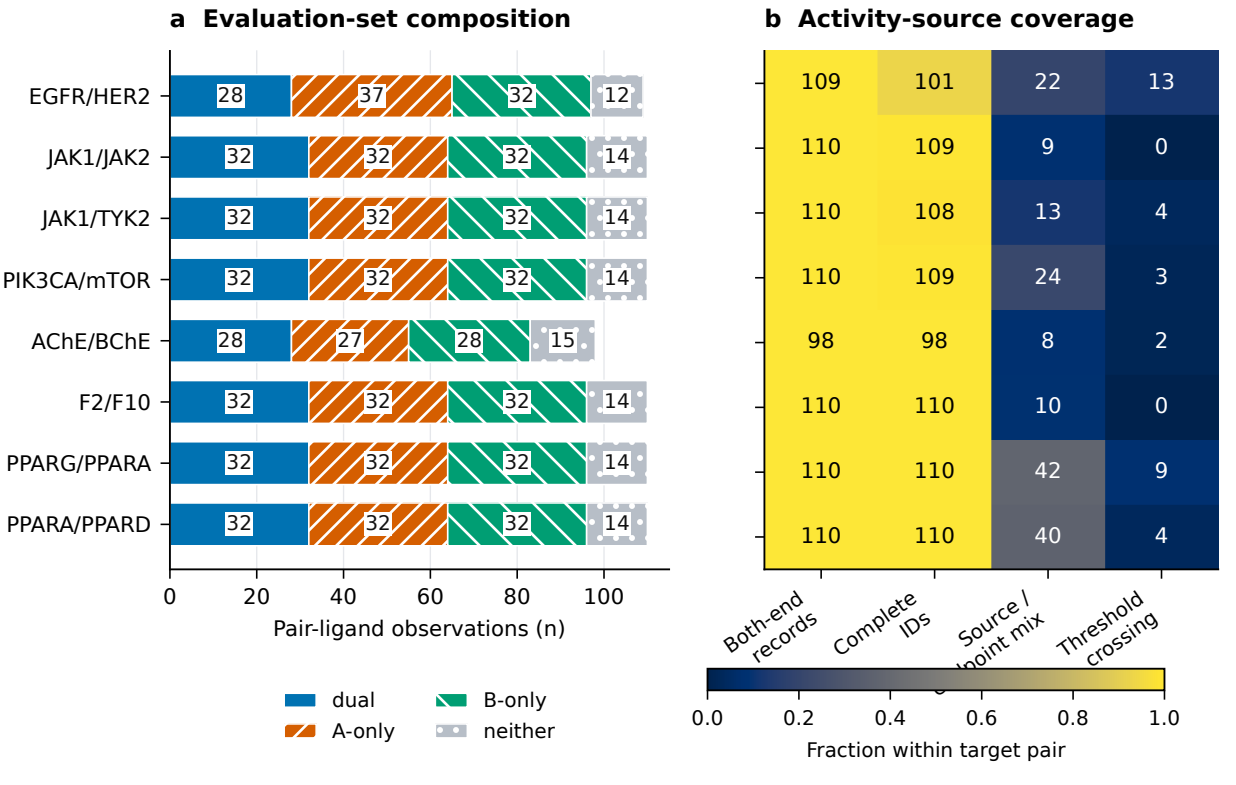
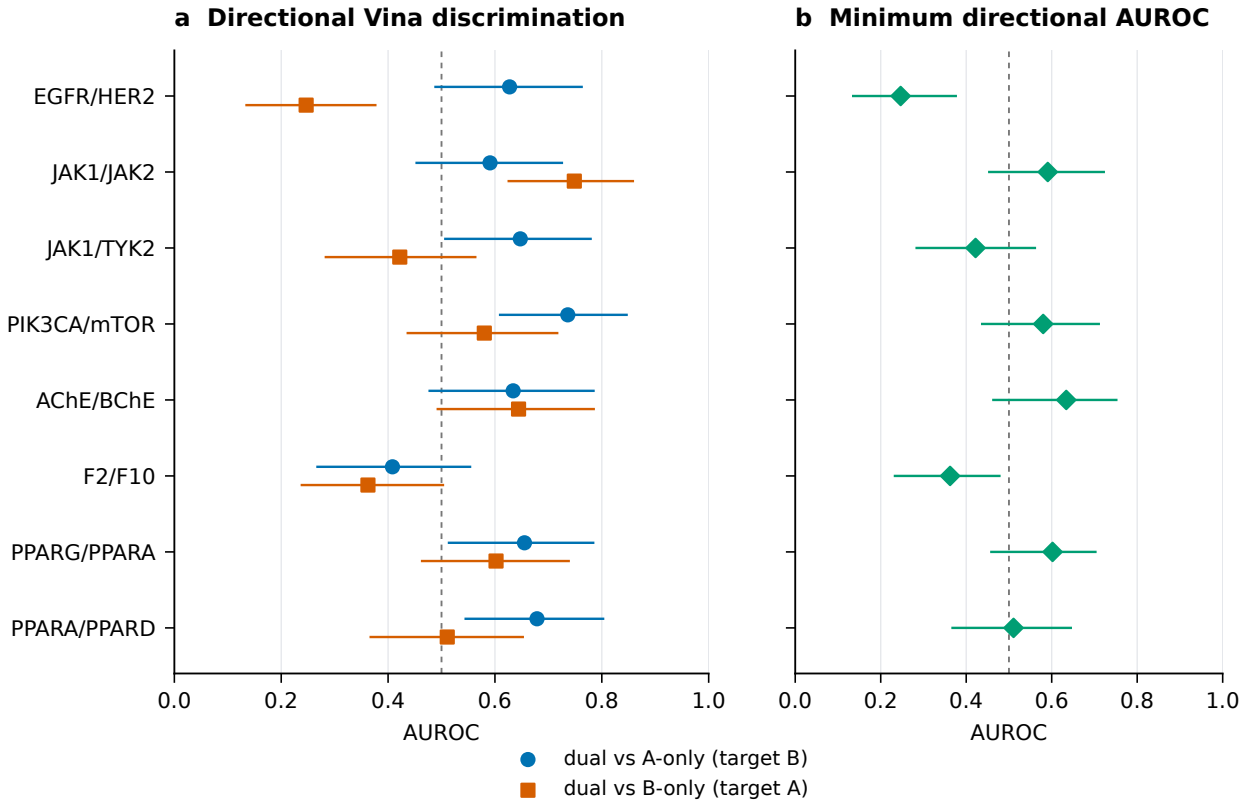
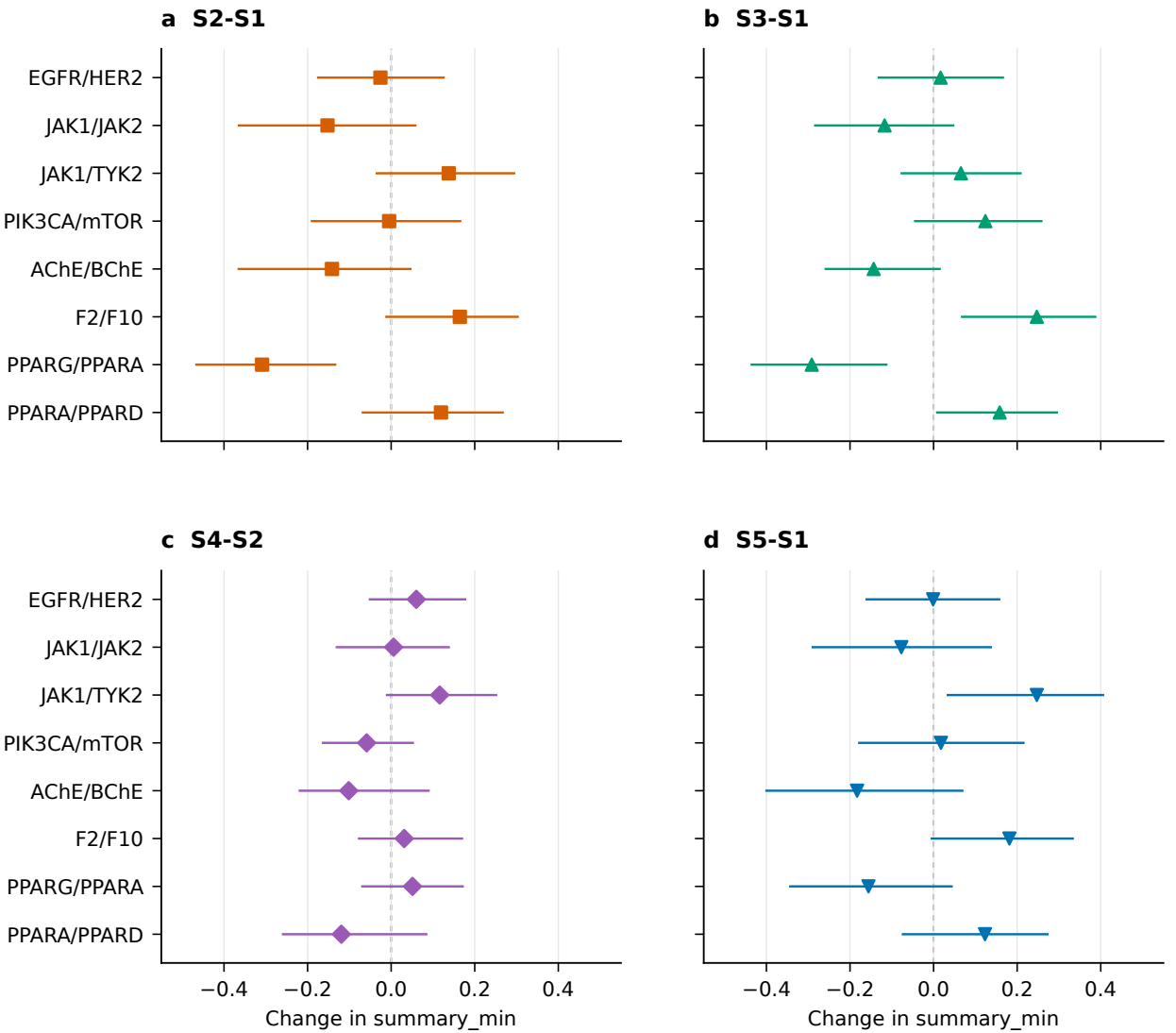
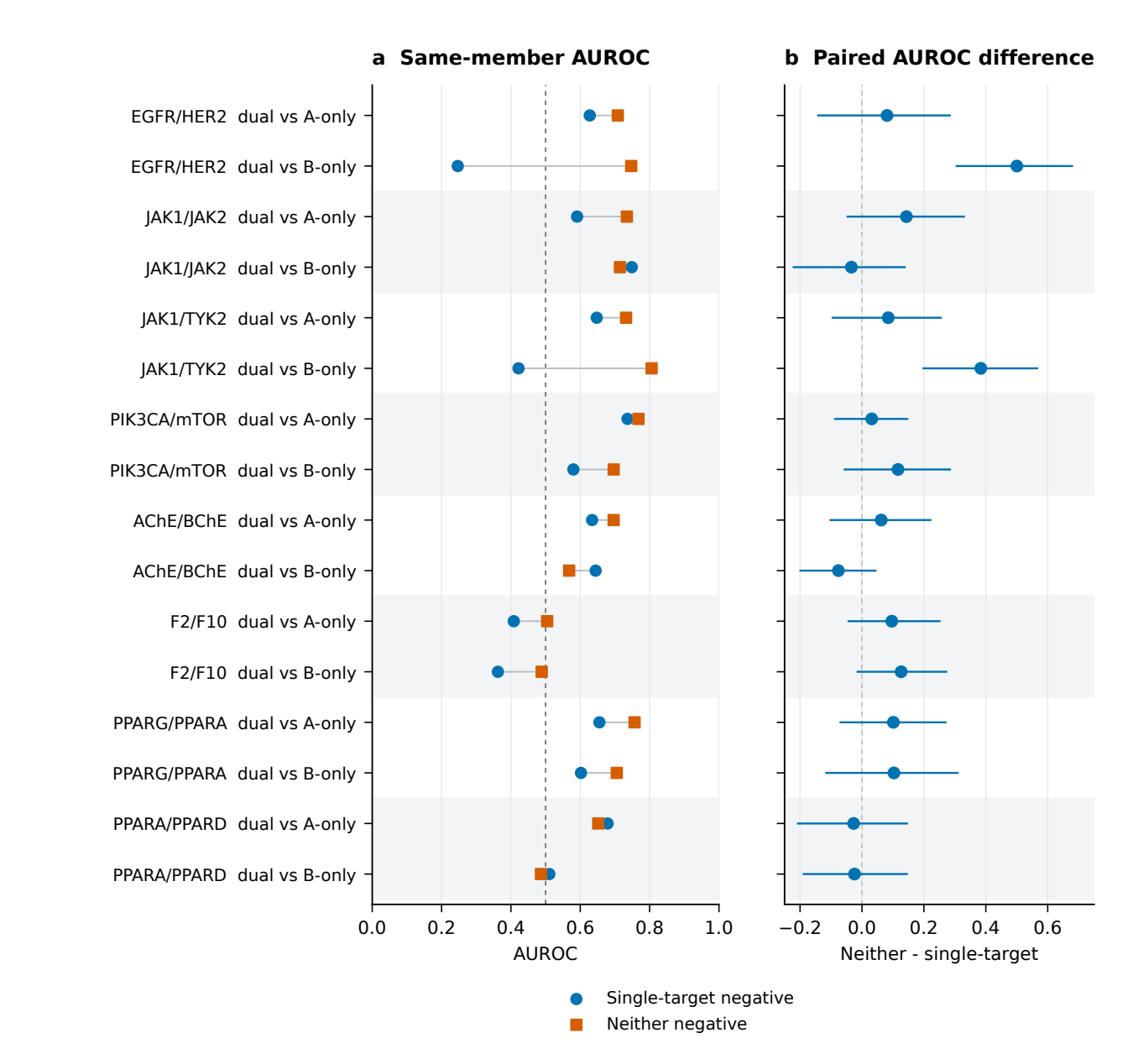
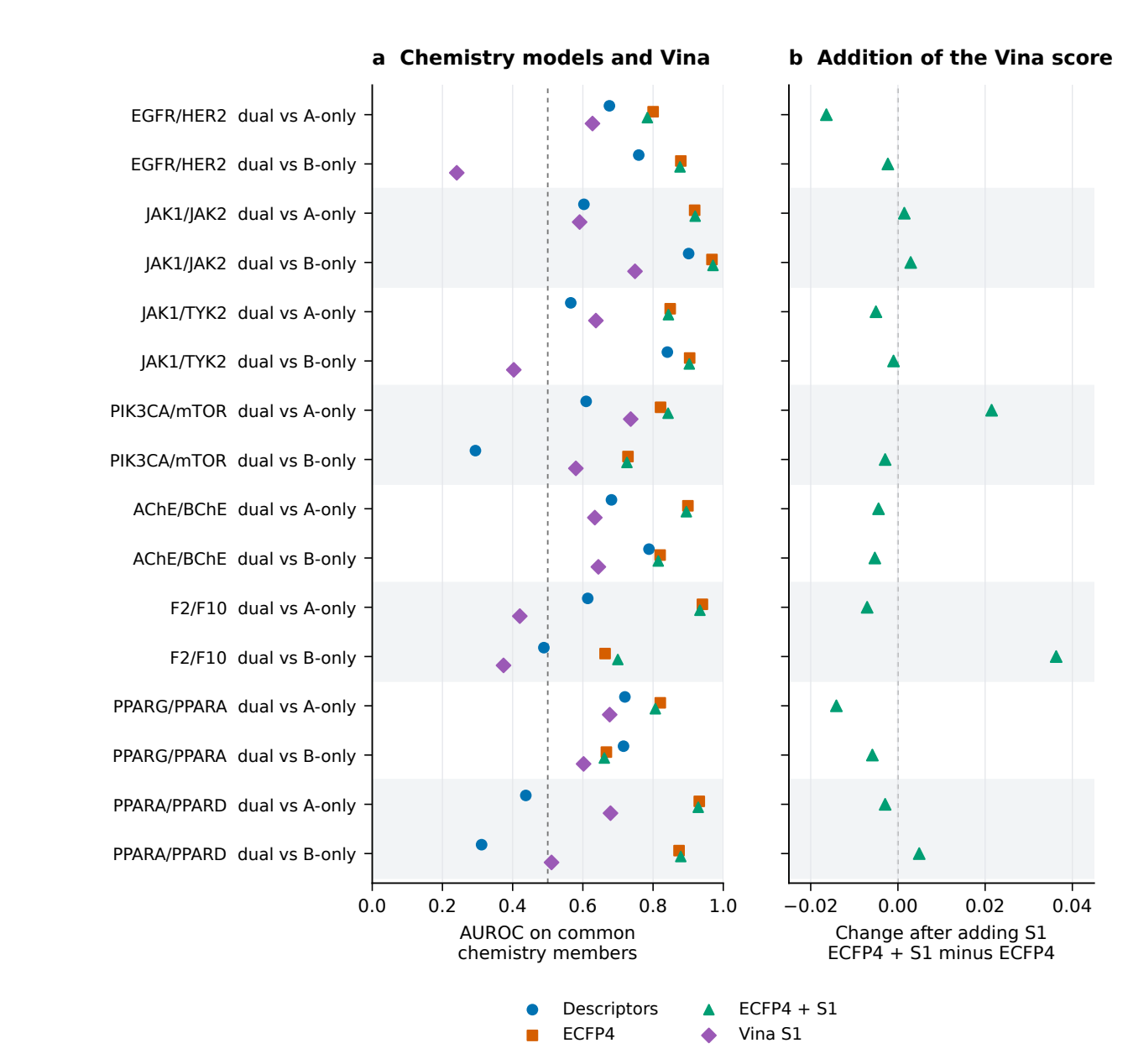
867 pair-ligand observations; 845 distinct ligand entities

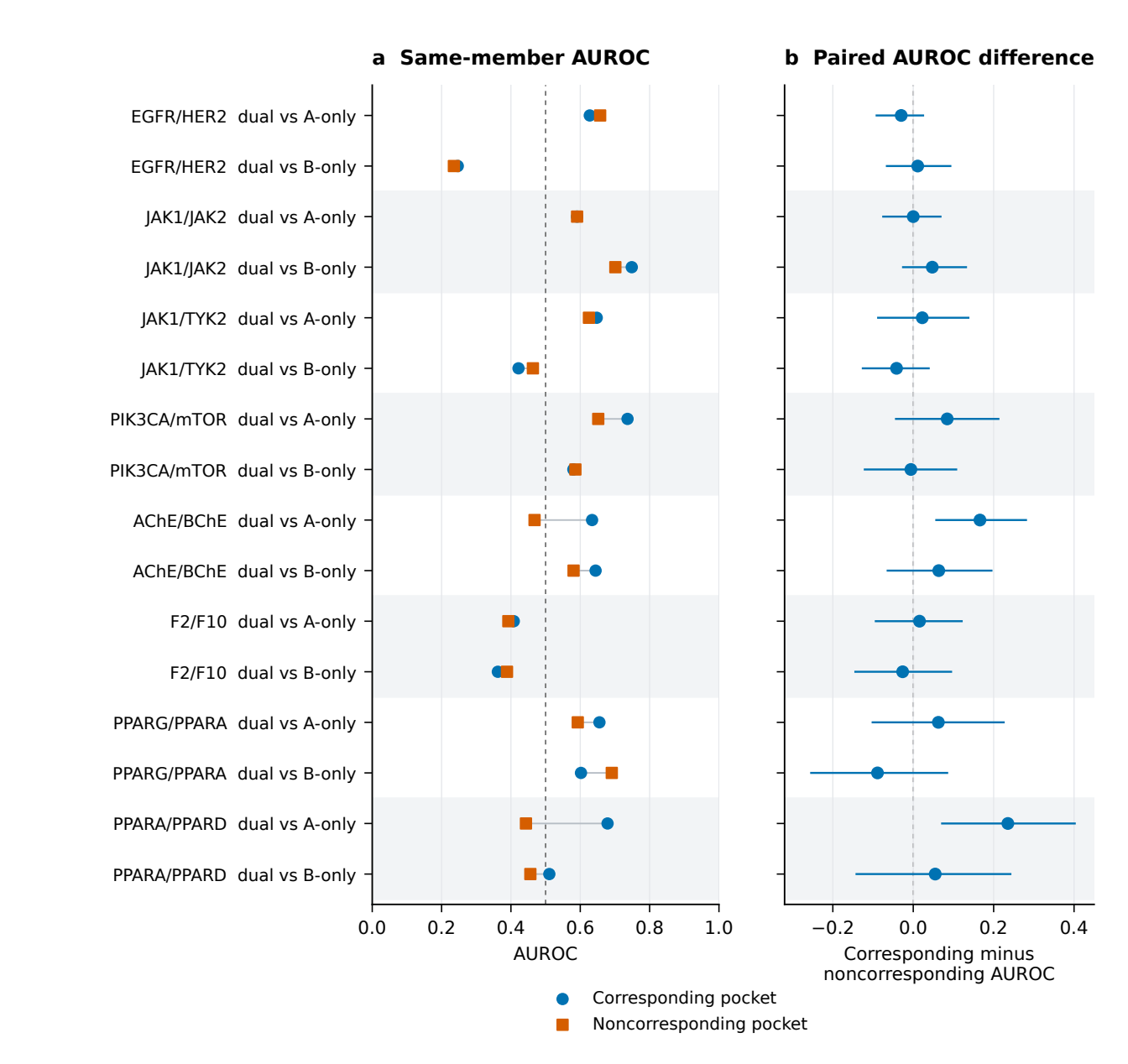


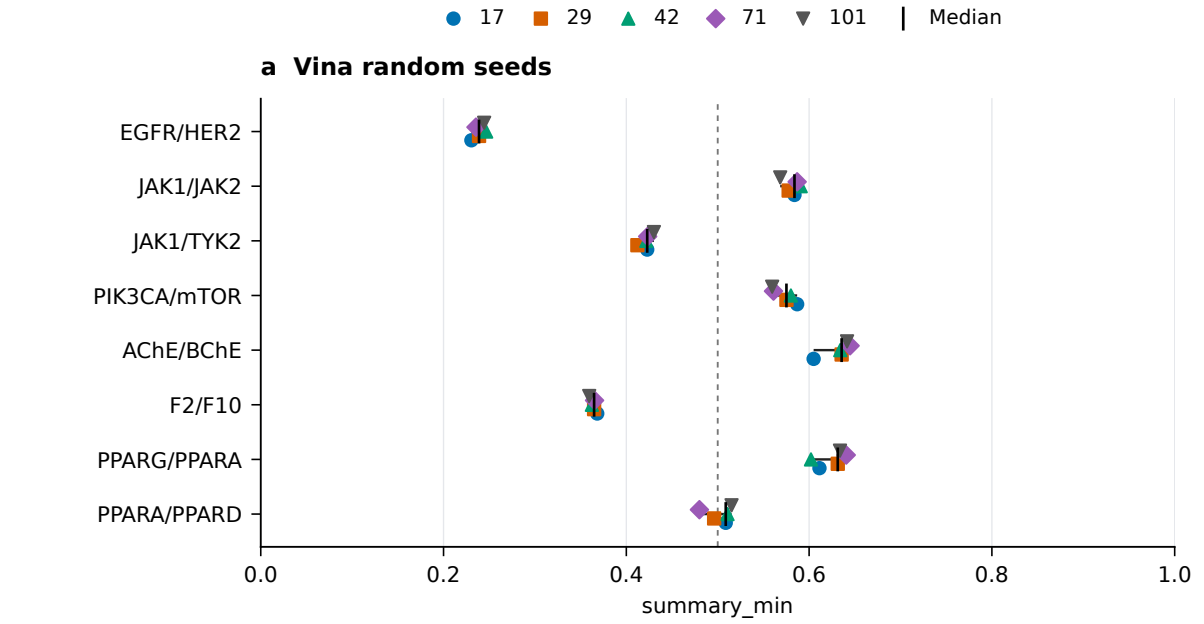

a Directional Vina discrimination
b Minimum directional AUROC
EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
0.0
0.2
0.4
0.6
0.8
1.0
AUROC
dual vs A-only (target B)
dual vs B-only (target A)

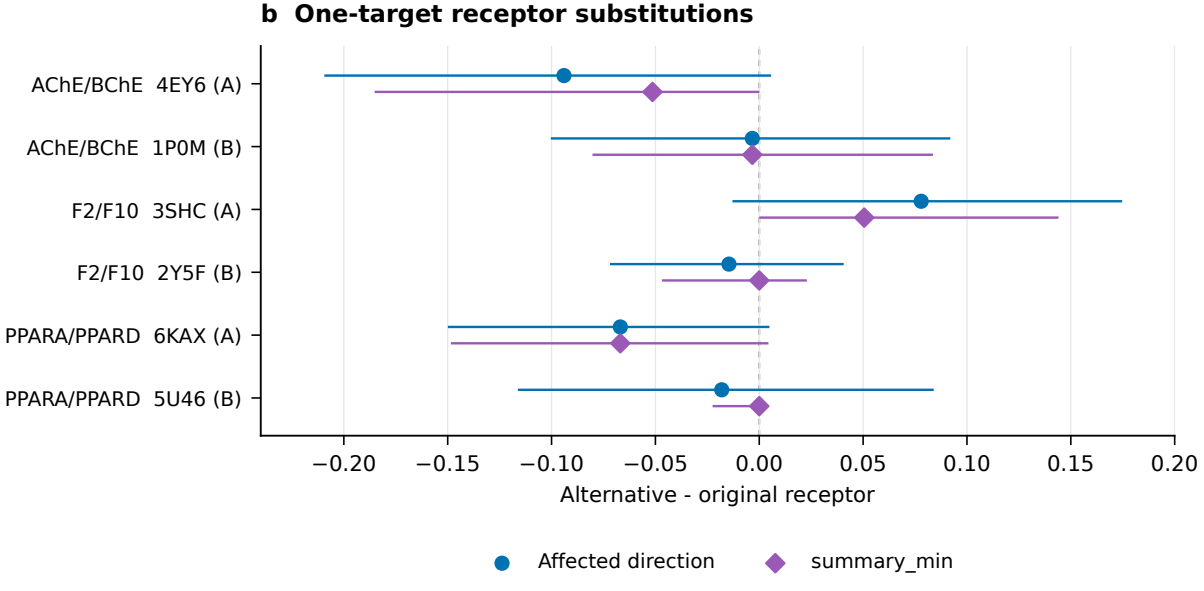

a S2-S1
b S3-S1
c S4-S2
d S5-S1
EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
−0.4
−0.2
0.0
0.2
0.4
Change in summary_min

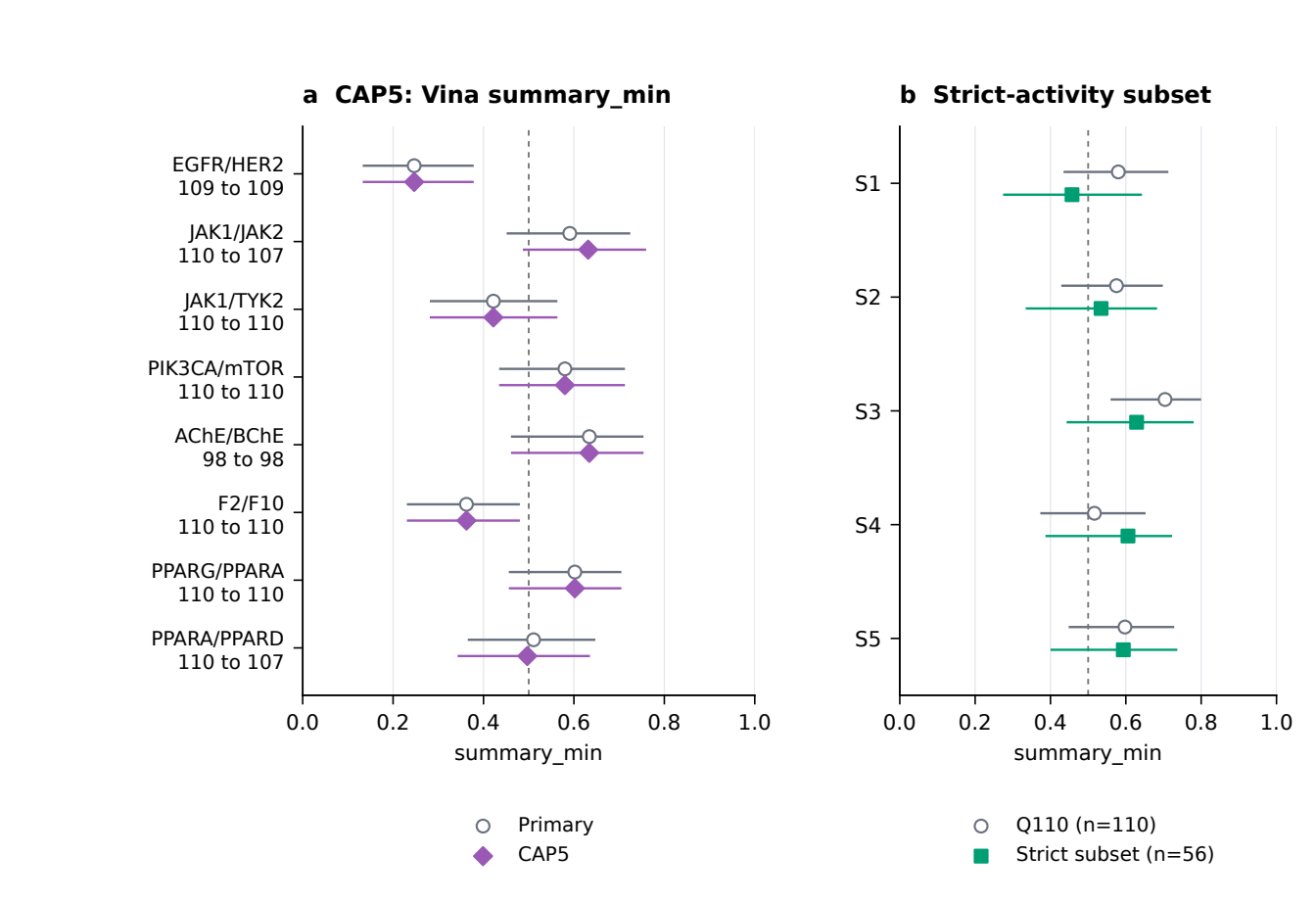

a Same-member AUROC
b Paired AUROC difference
EGFR/HER2 dual vs A-only
EGFR/HER2 dual vs B-only
JAK1/JAK2 dual vs A-only
JAK1/JAK2 dual vs B-only
JAK1/TYK2 dual vs A-only
JAK1/TYK2 dual vs B-only
PIK3CA/mTOR dual vs A-only
PIK3CA/mTOR dual vs B-only
AChE/BChE dual vs A-only
AChE/BChE dual vs B-only
F2/F10 dual vs A-only
F2/F10 dual vs B-only
PPARG/PPARA dual vs A-only
PPARG/PPARA dual vs B-only
PPARA/PPARD dual vs A-only
PPARA/PPARD dual vs B-only
0.0
0.2
0.4
0.6
0.8
1.0
AUROC
−0.2
0.0
0.2
0.4
0.6
Neither - single-target
Single-target negative
Neither negative

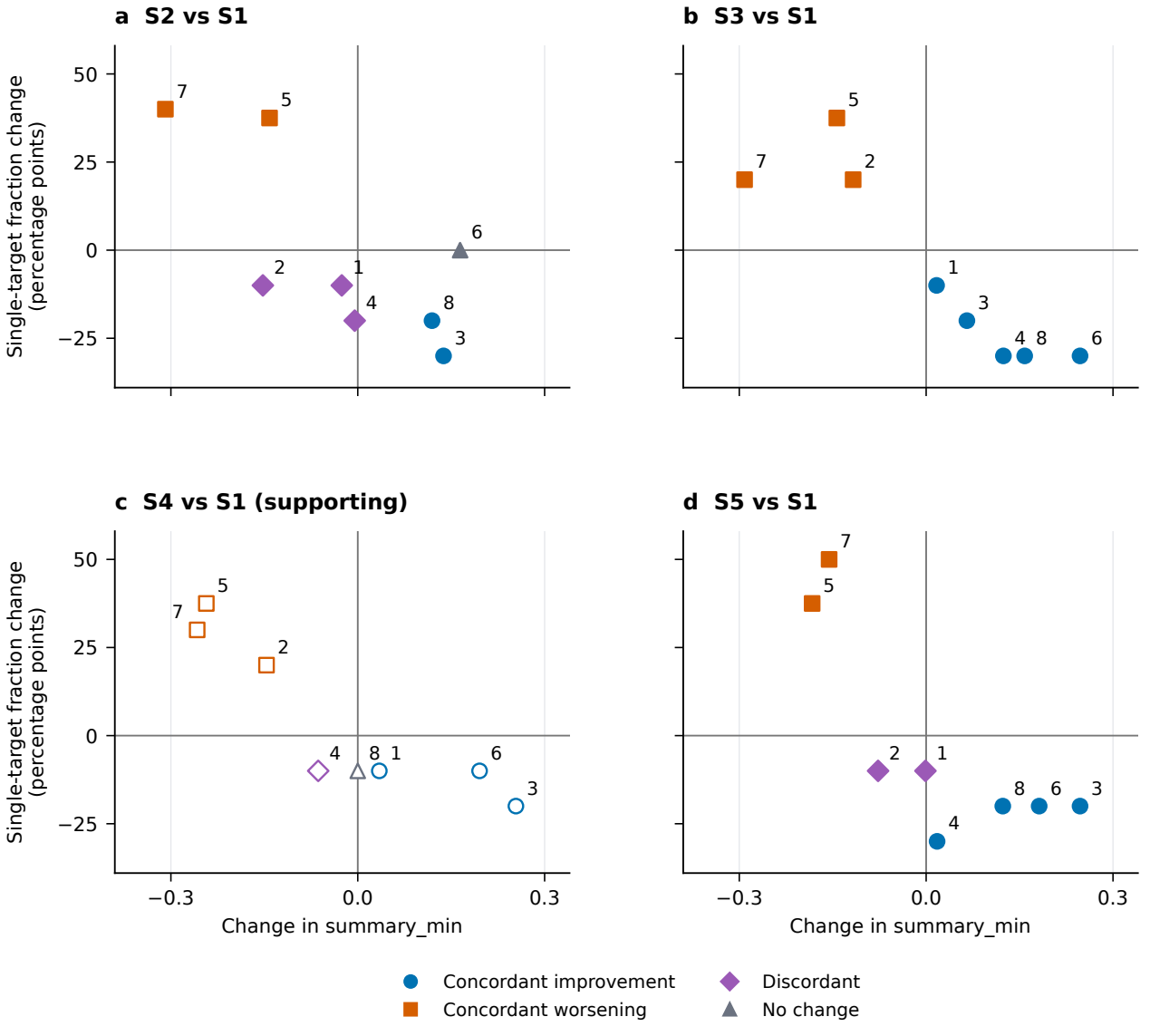

a Chemistry models and Vina
b Addition of the Vina score
EGFR/HER2 dual vs A-only
EGFR/HER2 dual vs B-only
JAK1/JAK2 dual vs A-only
JAK1/JAK2 dual vs B-only
JAK1/TYK2 dual vs A-only
JAK1/TYK2 dual vs B-only
PIK3CA/mTOR dual vs A-only
PIK3CA/mTOR dual vs B-only
AChE/BChE dual vs A-only
AChE/BChE dual vs B-only
F2/F10 dual vs A-only
F2/F10 dual vs B-only
PPARG/PPARA dual vs A-only
PPARG/PPARA dual vs B-only
PPARA/PPARD dual vs A-only
PPARA/PPARD dual vs B-only
0.0
0.2
0.4
0.6
0.8
1.0
AUROC on common chemistry members
−0.02
0.00
0.02
0.04
Change after adding S1 ECFP4 + S1 minus ECFP4
Descriptors
ECFP4
ECFP4 + S1
Vina S1

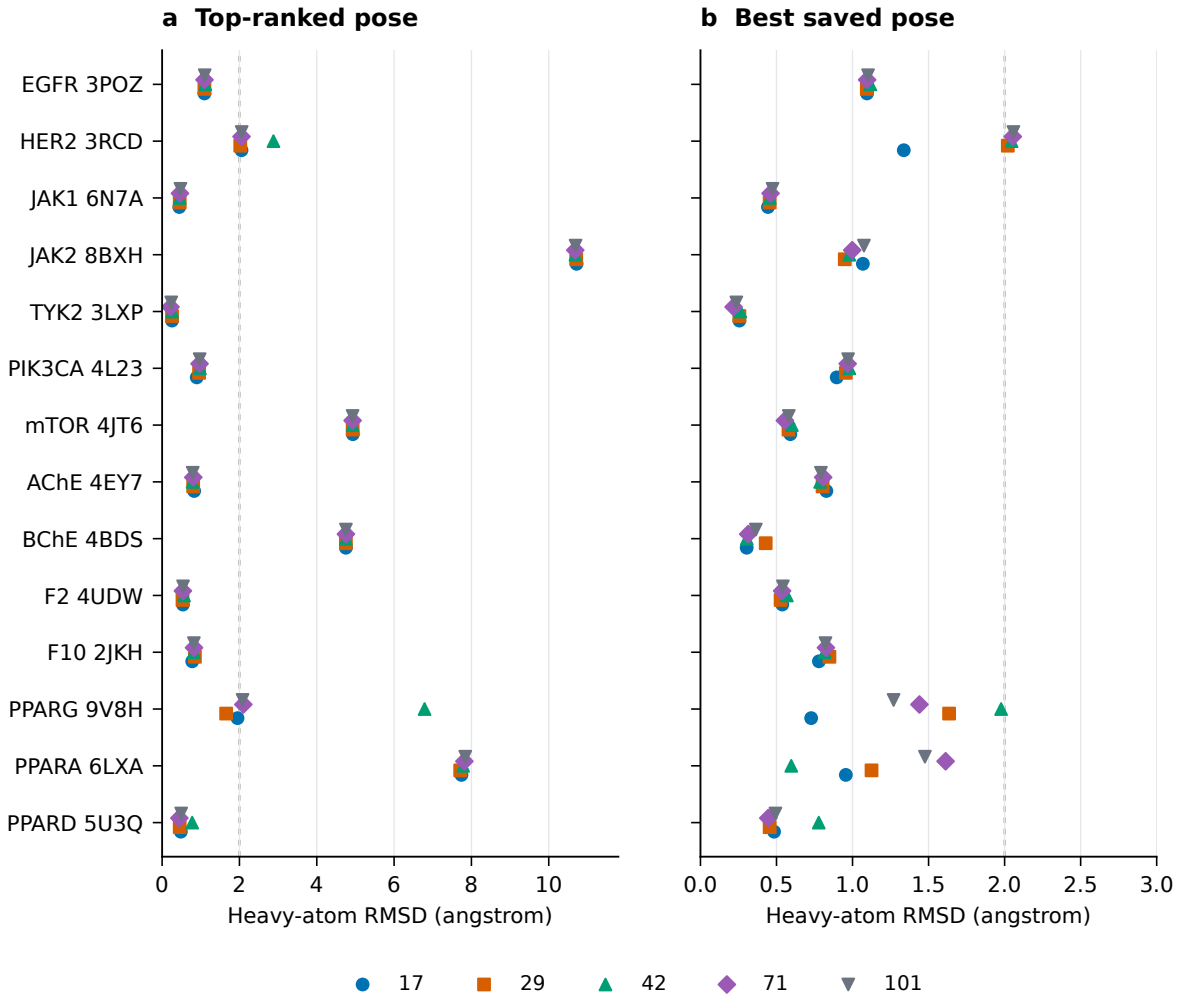

a Same-member AUROC
b Paired AUROC difference
EGFR/HER2 dual vs A-only
EGFR/HER2 dual vs B-only
JAK1/JAK2 dual vs A-only
JAK1/JAK2 dual vs B-only
JAK1/TYK2 dual vs A-only
JAK1/TYK2 dual vs B-only
PIK3CA/mTOR dual vs A-only
PIK3CA/mTOR dual vs B-only
AChE/BChE dual vs A-only
AChE/BChE dual vs B-only
F2/F10 dual vs A-only
F2/F10 dual vs B-only
PPARG/PPARA dual vs A-only
PPARG/PPARA dual vs B-only
PPARA/PPARD dual vs A-only
PPARA/PPARD dual vs B-only
0.0
0.2
0.4
0.6
0.8
1.0
AUROC
−0.2
0.0
0.2
0.4
Corresponding minus
noncorresponding AUROC
Corresponding pocket
Noncorresponding pocket


17
29
42
71
101
Median
a Vina random seeds
EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
0.0
0.2
0.4
0.6
0.8
1.0
summary_min


b One-target receptor substitutions
AChE/BChE 4EY6 (A)
AChE/BChE 1P0M (B)
F2/F10 3SHC (A)
F2/F10 2Y5F (B)
PPARA/PPARD 6KAX (A)
PPARA/PPARD 5U46 (B)
−0.20
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
0.20
Alternative - original receptor
Affected direction
summary_min


a CAP5: Vina summary_min
EGFR/HER2
109 to 109
JAK1/JAK2
110 to 107
JAK1/TYK2
110 to 110
PIK3CA/mTOR
110 to 110
AChE/BChE
98 to 98
F2/F10
110 to 110
PPARG/PPARA
110 to 110
PPARA/PPARD
110 to 107
0.0
0.2
0.4
0.6
0.8
1.0
summary_min
Primary
CAP5
b Strict-activity subset
S1
S2
S3
S4
S5
0.0
0.2
0.4
0.6
0.8
1.0
summary_min
Q110 (n=110)
Strict subset (n=56)


a S2 vs S1
b S3 vs S1
c S4 vs S1 (supporting)
d S5 vs S1
Single-target fraction change (percentage points)
Change in summary_min
50
25
0
−25
−0.3
0.0
0.3
Concordant improvement
Concordant worsening
Discordant
No change
1 EGFR/HER2
2 JAK1/JAK2
3 JAK1/TYK2
4 PIK3CA/mTOR
5 AChE/BChE
6 F2/F10
7 PPARG/PPARA
8 PPARA/PPARD


a Top-ranked pose
b Best saved pose
EGFR 3POZ
HER2 3RCD
JAK1 6N7A
JAK2 8BXH
TYK2 3LXP
PIK3CA 4L23
mTOR 4JT6
AChE 4EY7
BChE 4BDS
F2 4UDW
F10 2JKH
PPARG 9V8H
PPARA 6LXA
PPARD 5U3Q
0
2
4
6
8
10
0.0
0.5
1.0
1.5
2.0
2.5
3.0
Heavy-atom RMSD (angstrom)
Heavy-atom RMSD (angstrom)
17
29
42
71
101

**a dual, target A**

| | S1 | S2 | S3 | S4 | S5 |
|---|---|---|---|---|---|
| 1 | 28/28 | 28/28 | 28/28 | 28/28 | 28/28 |
| 2 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 3 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 4 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 5 | 27/28 | 27/28 | 27/28 | 27/28 | 27/28 |
| 6 | 32/32 | 31/32 | 32/32 | 31/32 | 31/32 |
| 7 | 32/32 | 30/32 | 32/32 | 30/32 | 32/32 |
| 8 | 32/32 | 27/32 | 32/32 | 27/32 | 32/32 |

**b dual, target B**

| | S1 | S2 | S3 | S4 | S5 |
|---|---|---|---|---|---|
| 1 | 28/28 | 28/28 | 28/28 | 28/28 | 28/28 |
| 2 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 3 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 4 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 5 | 27/28 | 27/28 | 27/28 | 27/28 | 27/28 |
| 6 | 32/32 | 32/32 | 32/32 | 32/32 | 31/32 |
| 7 | 32/32 | 30/32 | 32/32 | 30/32 | 32/32 |
| 8 | 32/32 | 27/32 | 32/32 | 27/32 | 32/32 |

**c A-only, target A**

| | S1 | S2 | S3 | S4 | S5 |
|---|---|---|---|---|---|
| 1 | 37/37 | 37/37 | 37/37 | 37/37 | 37/37 |
| 2 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 3 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 4 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 5 | 25/27 | 25/27 | 25/27 | 25/27 | 27/27 |
| 6 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 7 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 8 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |

**d A-only, target B**

| | S1 | S2 | S3 | S4 | S5 |
|---|---|---|---|---|---|
| 1 | 37/37 | 37/37 | 37/37 | 37/37 | 37/37 |
| 2 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 3 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 4 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 5 | 25/27 | 25/27 | 25/27 | 25/27 | 27/27 |
| 6 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 7 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 8 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |

**e B-only, target A**

| | S1 | S2 | S3 | S4 | S5 |
|---|---|---|---|---|---|
| 1 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 2 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 3 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 4 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 5 | 28/28 | 27/28 | 28/28 | 27/28 | 28/28 |
| 6 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 7 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 8 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |

**f B-only, target B**

| | S1 | S2 | S3 | S4 | S5 |
|---|---|---|---|---|---|
| 1 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 2 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 3 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 4 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 5 | 28/28 | 27/28 | 28/28 | 27/28 | 28/28 |
| 6 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 7 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |
| 8 | 32/32 | 32/32 | 32/32 | 32/32 | 32/32 |

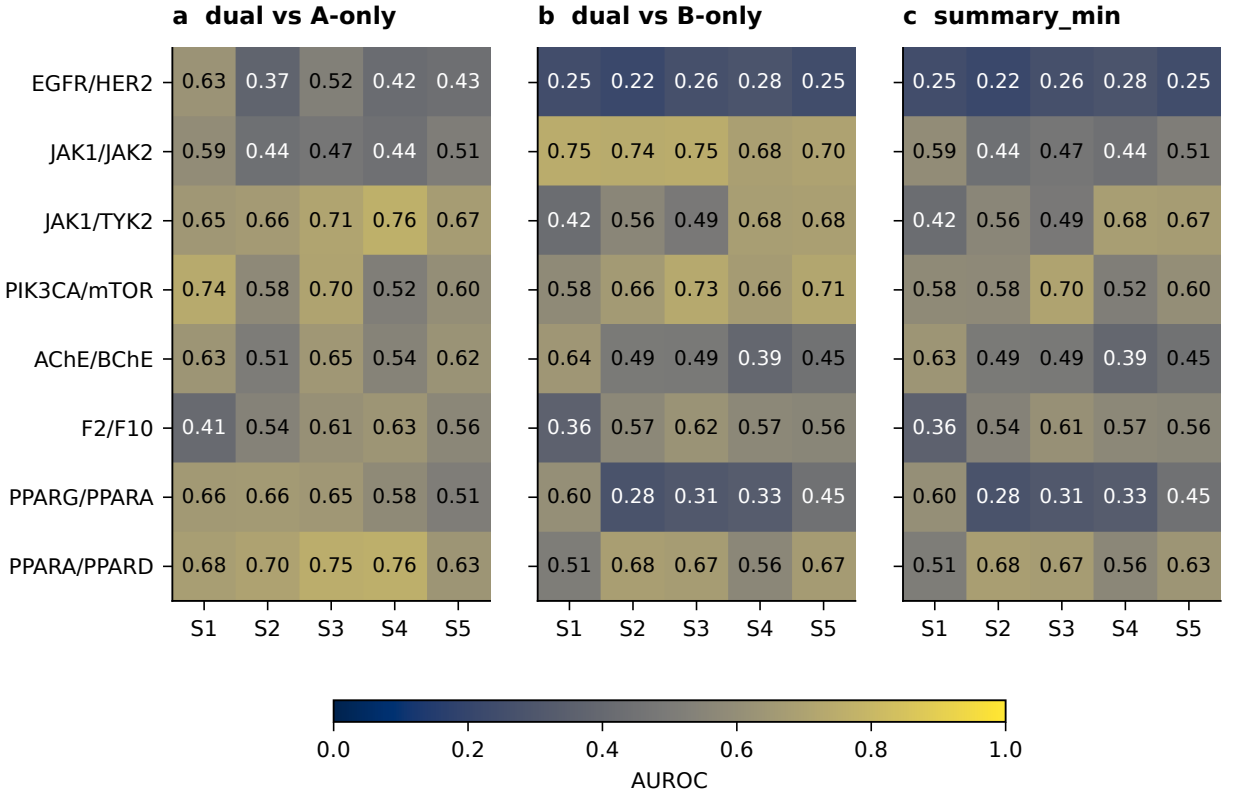
**g neither, target A**

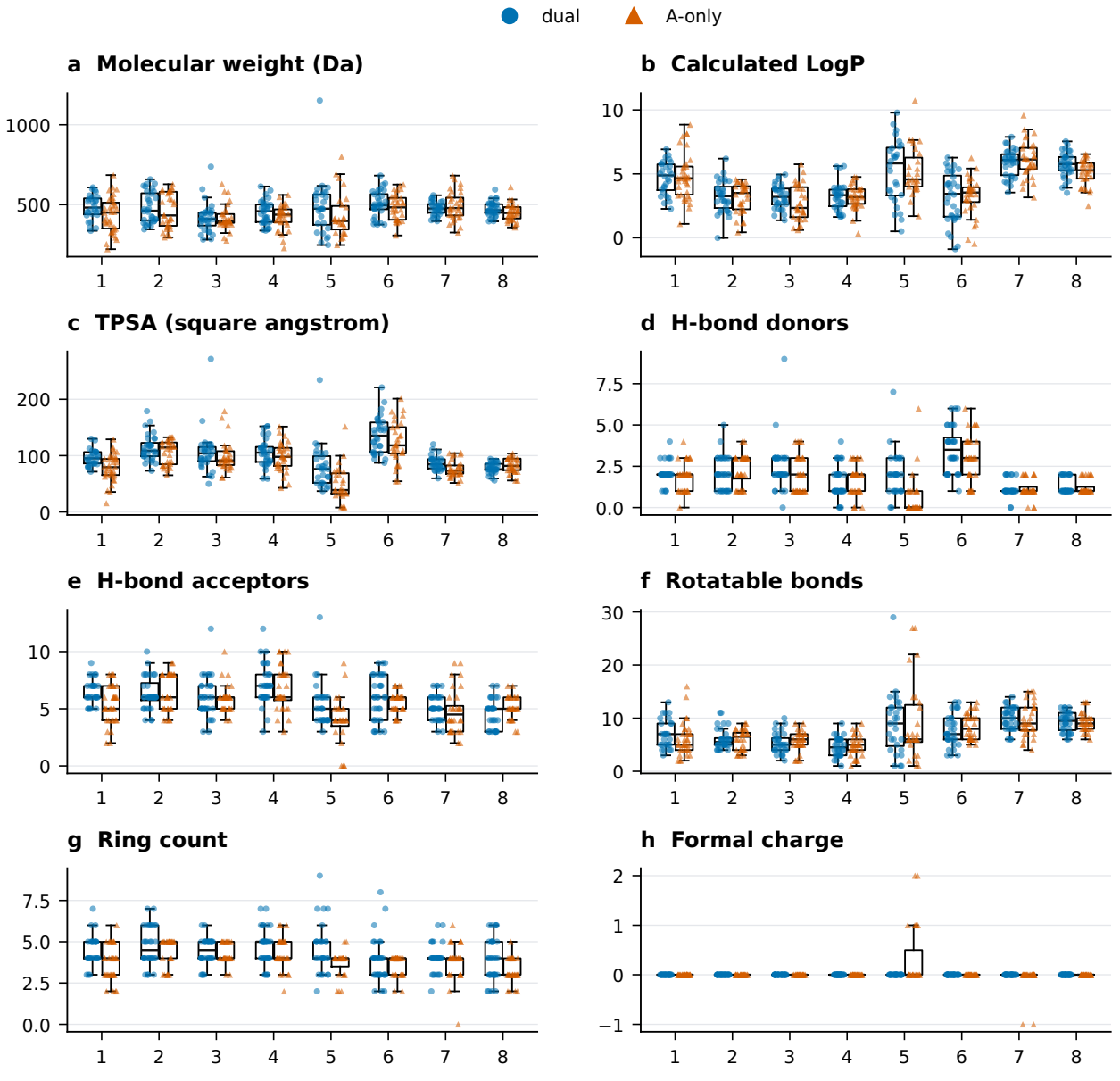
| | S1 | S2 | S3 | S4 | S5 |
|---|---|---|---|---|---|
| 1 | 12/12 | 12/12 | 12/12 | 12/12 | 12/12 |
| 2 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 |
| 3 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 |
| 4 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 |
| 5 | 15/15 | 15/15 | 15/15 | 15/15 | 15/15 |
| 6 | 14/14 | 14/14 | 14/14 | 14/14 | 13/14 |
| 7 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 |
| 8 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 |

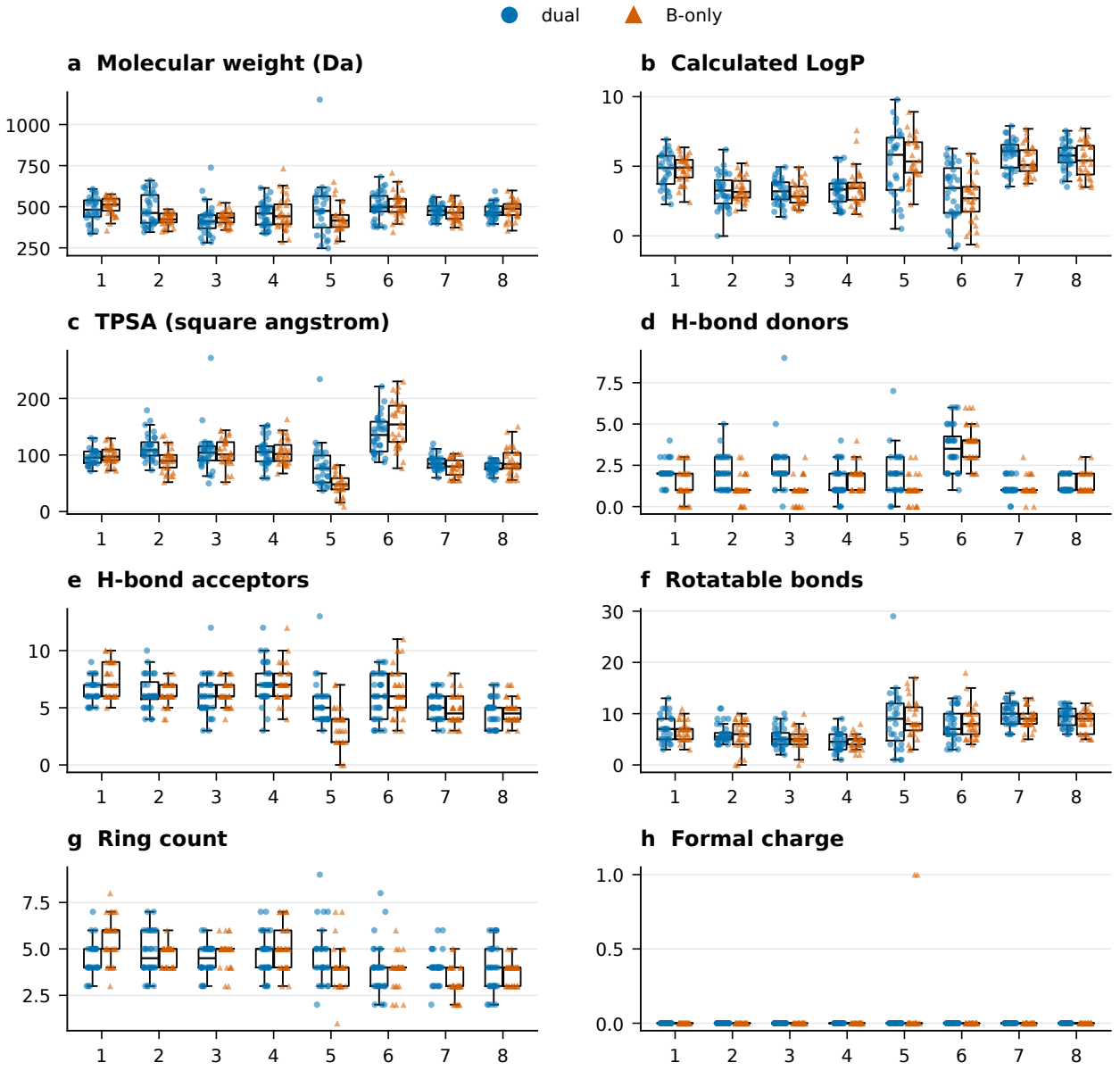
**h neither, target B**

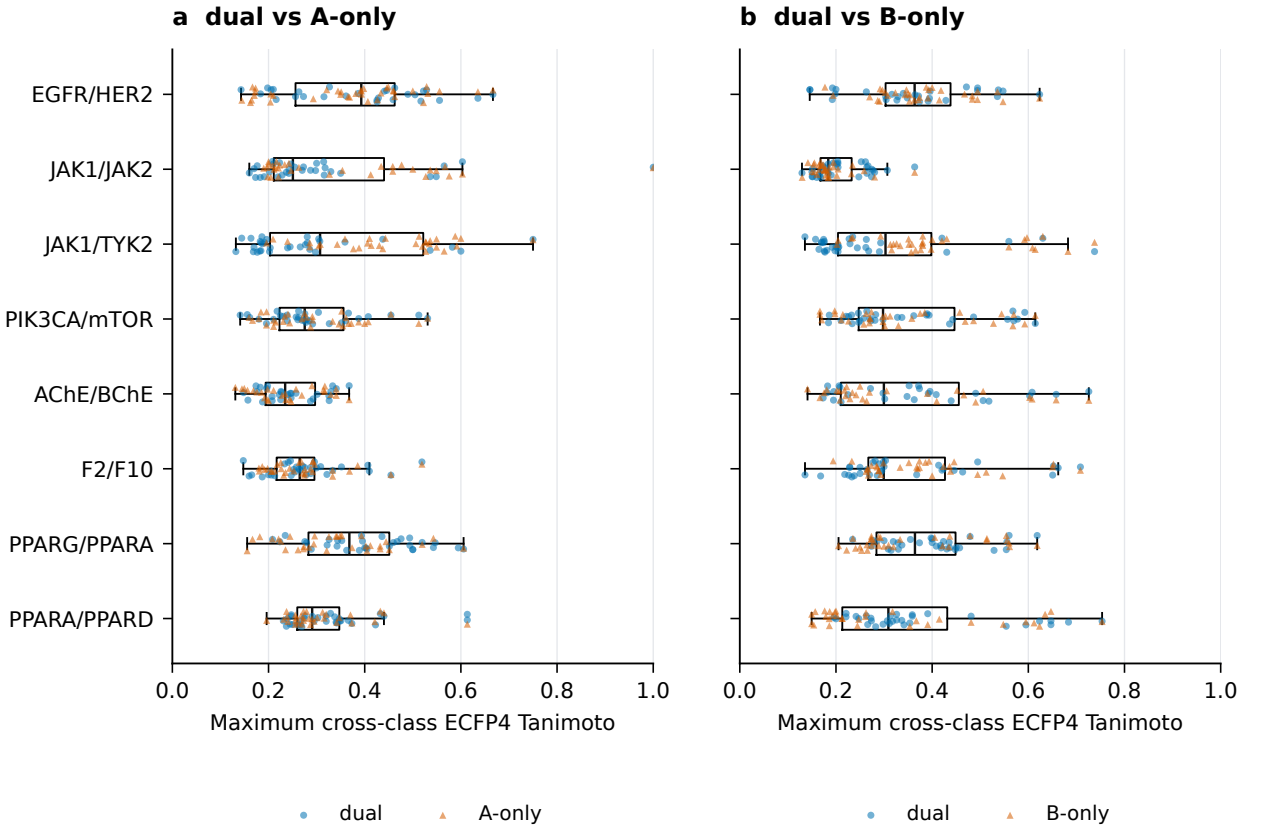
| | S1 | S2 | S3 | S4 | S5 |
|---|---|---|---|---|---|
| 1 | 12/12 | 12/12 | 12/12 | 12/12 | 12/12 |
| 2 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 |
| 3 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 |
| 4 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 |
| 5 | 15/15 | 15/15 | 15/15 | 15/15 | 15/15 |
| 6 | 14/14 | 14/14 | 14/14 | 14/14 | 13/14 |
| 7 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 |
| 8 | 14/14 | 14/14 | 14/14 | 14/14 | 14/14 |

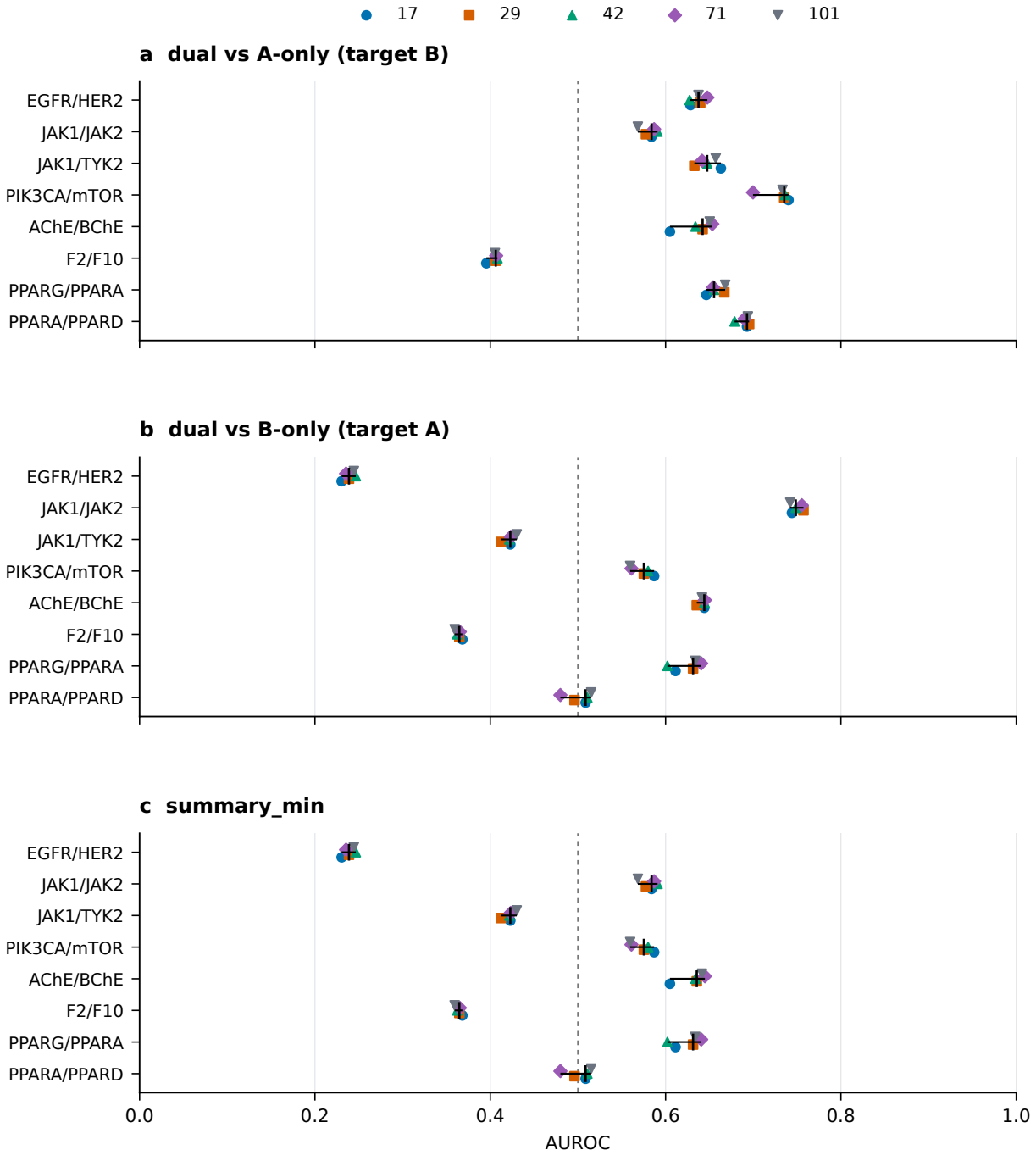
1 EGFR/HER2 2 JAK1/JAK2 3 JAK1/TYK2 4 PIK3CA/mTOR
5 AChE/BChE 6 F2/F10 7 PPARG/PPARA 8 PPARA/PPARD

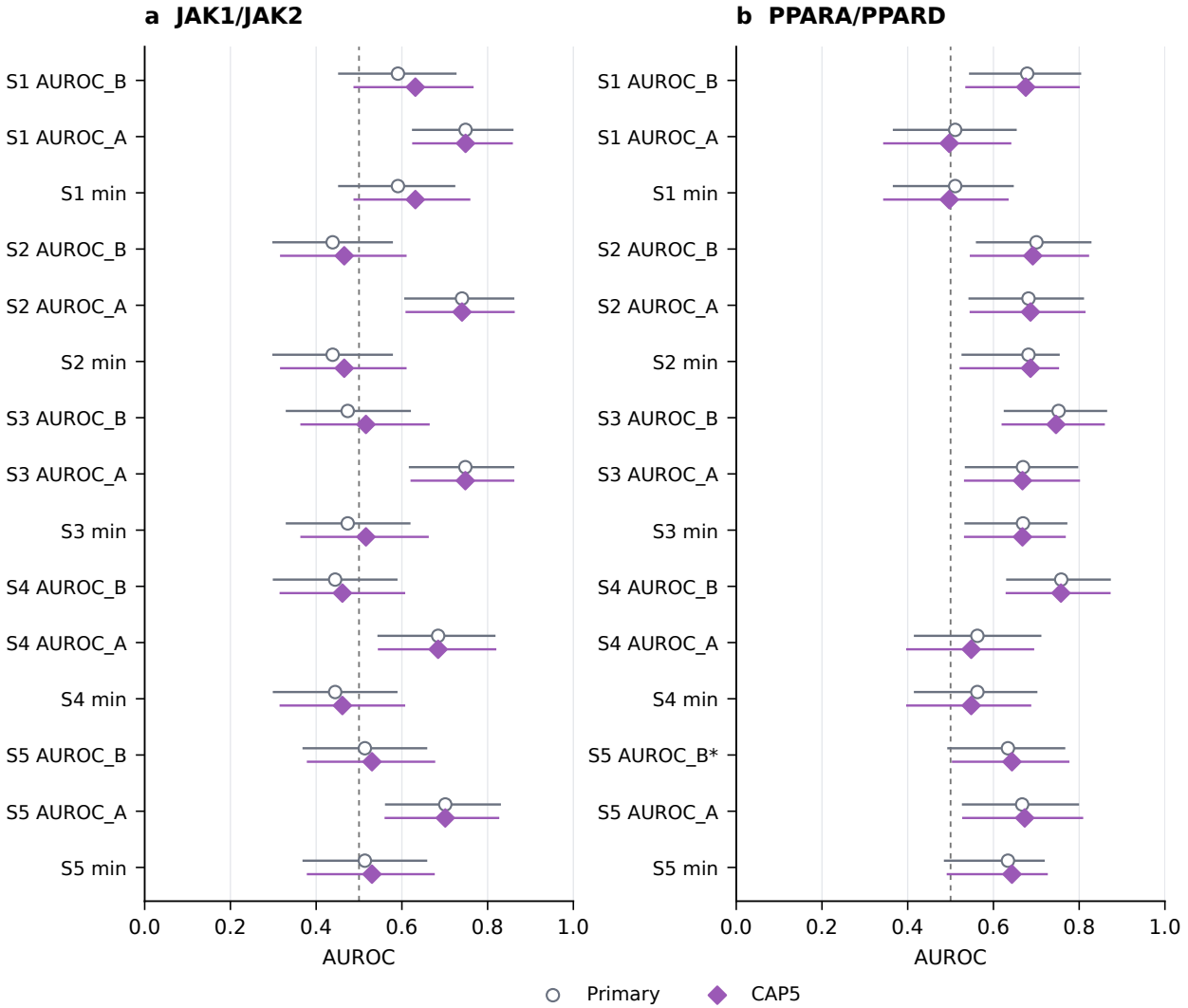

a dual vs A-only
b dual vs B-only
c summary_min
EGFR/HER2 0.63 0.37 0.52 0.42 0.43 0.25 0.22 0.26 0.28 0.25 0.25 0.22 0.26 0.28 0.25
JAK1/JAK2 0.59 0.44 0.47 0.44 0.51 0.75 0.74 0.75 0.68 0.70 0.59 0.44 0.47 0.44 0.51
JAK1/TYK2 0.65 0.66 0.71 0.76 0.67 0.42 0.56 0.49 0.68 0.68 0.42 0.56 0.49 0.68 0.67
PIK3CA/mTOR 0.74 0.58 0.70 0.52 0.60 0.58 0.66 0.73 0.66 0.71 0.58 0.58 0.70 0.52 0.60
AChE/BChE 0.63 0.51 0.65 0.54 0.62 0.64 0.49 0.49 0.39 0.45 0.63 0.49 0.49 0.39 0.45
F2/F10 0.41 0.54 0.61 0.63 0.56 0.36 0.57 0.62 0.57 0.56 0.36 0.54 0.61 0.57 0.56
PPARG/PPARA 0.66 0.66 0.65 0.58 0.51 0.60 0.28 0.31 0.33 0.45 0.60 0.28 0.31 0.33 0.45
PPARA/PPARD 0.68 0.70 0.75 0.76 0.63 0.51 0.68 0.67 0.56 0.67 0.51 0.68 0.67 0.56 0.63
S1 S2 S3 S4 S5
S1 S2 S3 S4 S5
S1 S2 S3 S4 S5
0.0 0.2 0.4 0.6 0.8 1.0
AUROC


dual
A-only
a Molecular weight (Da)
b Calculated LogP
c TPSA (square angstrom)
d H-bond donors
e H-bond acceptors
f Rotatable bonds
g Ring count
h Formal charge
1 EGFR/HER2 2 JAK1/JAK2 3 JAK1/TYK2 4 PIK3CA/mTOR
5 AChE/BChE 6 F2/F10 7 PPARG/PPARA 8 PPARA/PPARD

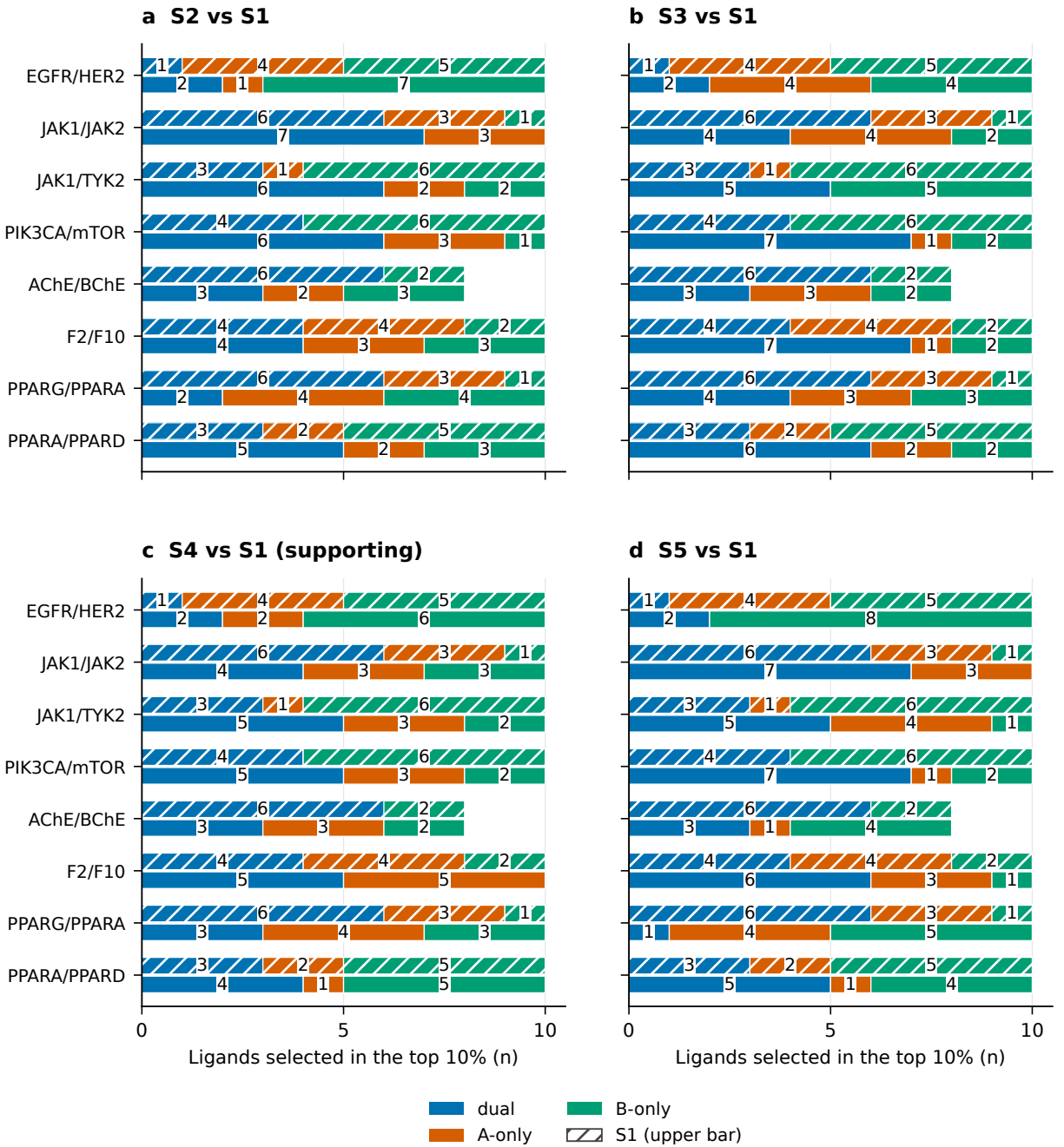

dual
B-only
a Molecular weight (Da)
b Calculated LogP
c TPSA (square angstrom)
d H-bond donors
e H-bond acceptors
f Rotatable bonds
g Ring count
h Formal charge
1 EGFR/HER2 2 JAK1/JAK2 3 JAK1/TYK2 4 PIK3CA/mTOR
5 AChE/BChE 6 F2/F10 7 PPARG/PPARA 8 PPARA/PPARD


a dual vs A-only
b dual vs B-only
EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
0.0
0.2
0.4
0.6
0.8
1.0
Maximum cross-class ECFP4 Tanimoto
dual
A-only
B-only


17
29
42
71
101
a dual vs A-only (target B)
b dual vs B-only (target A)
c summary_min
EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
0.0
0.2
0.4
0.6
0.8
1.0
AUROC




* PPARA/PPARD S5 AUROC_B: lower 95% CI 0.4922 to 0.5030


a S2 vs S1
b S3 vs S1
c S4 vs S1 (supporting)
d S5 vs S1
EGFR/HER2
JAK1/JAK2
JAK1/TYK2
PIK3CA/mTOR
AChE/BChE
F2/F10
PPARG/PPARA
PPARA/PPARD
0
5
10
Ligands selected in the top 10% (n)
dual
A-only
B-only
S1 (upper bar)

## Scoring and pose-selection workflows

